SIMATS ENGINEERING
ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
QUESTIONS: 5 **Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Tejaswi Reddy K(192371036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Tejaswi Reddy.K

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause (□) if the goal is proved.

SOLUTION –

Step 1: Represent in Propositional Logic

Let:

- **R** = It is raining.
- **W** = The ground is wet.

Statement 1: **R → W**

Statement 2: **R**

Goal: **W**

Step 2: Convert to CNF

Implication:

R → W ≡ ¬R ∨ W

Therefore, the clauses are:

- **C1: ¬R ∨ W**
- **C2: R**

Step 3: Negate the Goal

Goal: **W**

Negated goal: **¬W**

So:

- **C3: ¬W**

Step 4: Apply Resolution

Resolve C1 and C2:

C1: ¬R ∨ W
C2: R

Resolving **¬R** with **R**:

W

New clause:

C4: W

Now resolve C4 with the negated goal C3:

C4: W
C3: ¬W

Therefore: □ **(Empty Clause)**

The empty clause □ is derived. Therefore, the negation of the goal leads to a contradiction.

Hence, the goal W is proved.

□ **The ground is wet.**

Question 2 – Student Assignment Submission
Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks
1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.

2. Conclude whether the goal is proved.

SOLUTION –

Step 1: Propositional Logic

Let:

- **S** = Rahul submitted the assignment.
- **M** = Rahul receives internal marks.

Statement 1: **S → M**

Statement 2: **S**

Goal: **M**

Step 2: Convert to CNF

The implication: **S → M** is equivalent to: **¬S ∨ M**

Therefore:

- **C1: ¬S ∨ M**
- **C2: S**

Step 3: Negate the Goal

Goal: **M**

Negated goal: **¬M**

Therefore:

- **C3: ¬M**

Step 4: Apply Resolution

Resolve C1 and C2:

C1: ¬S ∨ M
C2: S

Resolving **¬S** and **S** gives: **M**

Therefore:

C4: M

Now resolve C4 with C3:

C4: M
C3: ¬M

This gives: □

Since the empty clause □ is derived, the negated goal is contradictory.

Therefore: □ **Rahul receives internal marks.**

Question 3 – Library Membership
Knowledge Base
1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks
1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

SOLUTION –

Step 1: Propositional Logic

Let:

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

Statement 1:

$\mathbf{L \rightarrow B}$

Statement 2:

L

Goal:

B

Step 2: Convert to CNF

The implication:

$\mathbf{L \rightarrow B}$

becomes:

¬L ∨ B

Therefore, the clauses are:

- **C1: ¬L ▯ B**
- **C2: L**

Step 3: Negate the Goal

Goal:

B

Negated goal:

¬B

Therefore:

- **C3: ¬B**

Step 4: Perform Resolution

Resolve C1 and C2:

C1: ¬L ∨ B
C2: L

Resolve **¬L** with **L**:

B

Therefore:

C4: B

Now resolve C4 with the negated goal:

C4: B
C3: ¬B

Result: □

The empty clause □ is obtained.

Therefore, the original goal is proved.

▯ **Priya can borrow books.**

Question 4 – Placement Eligibility
Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

SOLUTION –

Step 1: Propositional Logic

Let:

- **A** = Arun cleared the aptitude test.
- **E** = Arun is eligible for placement.

Statement 1:

$A \rightarrow E$

Statement 2:

A

Goal:

E

Step 2: Convert to CNF

Convert:

$A \rightarrow E$

Using:

$P \rightarrow Q \equiv \neg P \lor Q$

Therefore:

$\neg A \lor E$

The CNF clauses are:

- **C1: $\neg A \lor E$**
- **C2: A**

Step 3: Negate the Goal

Goal:

E

Negated goal:

¬E

Therefore:

- **C3: ¬E**

Step 4: Apply Resolution

Resolve C1 and C2:

C1: ¬A ∨ E
C2: A

Resolve **¬A** with **A**:

E

Therefore:

C4: E

Now resolve C4 with C3:

C4: E
C3: ¬E

Result: □

The empty clause □ is obtained, which means the negation of the goal produces a contradiction.

 Arun is eligible for placement.

Question 5 – Access Control System
Knowledge Base
1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.
Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks
1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (□).
5. Explain each resolution step and write the final conclusion.

SOLUTION –

Step 1: Propositional Logic

Let:

- **P** = User entered the correct password.
- **A** = User is authenticated.
- **G** = User is granted access.

The knowledge base becomes:

1. **P → A**
2. **A → G**
3. **P**

Goal: **G**

Step 2: Convert All Implications to CNF

For the first implication: **P → A** becomes: **¬P ∨ A**

Therefore:

C1: ¬P ∨ A

For the second implication: **A → G** becomes: **¬A ∨ G**

Therefore:

C2: ¬A ∨ G

The third statement is:

C3: P

Step 3: Negate the Goal

Goal: **G**

Negated goal: **¬G**

Therefore:

C4: ¬G

Our complete clause set is:

Clause	Expression
C1	¬P ∨ A
C2	¬A ∨ G
C3	P
C4	¬G

Step 4: First Resolution

Resolve C1 and C3:

C1: $\neg P \lor A$
C3: P

Resolve **$\neg P$** and **P**.

Result: **A**

Therefore:

C5: A

This means the user is authenticated.

Step 5: Second Resolution

Now resolve C2 and C5:

C2: $\neg A \lor G$
C5: A

Resolve **$\neg A$** and **A**.

Result: **G**

Therefore:

C6: G

This means the user is granted access.

Step 6: Final Resolution

Now resolve C6 with the negated goal C4:

C6: G
C4: $\neg G$

Resolving **G** and **$\neg G$** gives: □

Therefore, the empty clause is derived.

Since the resolution process produces the **empty clause** □, the negation of the goal is contradictory.

Therefore, the original goal is logically proven.

□ **The user is granted access.**

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation